Supplementary of Unsupervised Graph-level Anomaly Detection via Multi-granular Graph Structure Learning

Ge Zhang[1] , **Huimei Li**[1] , **Guohao Sun**[1*] , **Xiu Fang**[1] , **Xixun Lin**[2] , **Xiaobao Wang**[3] , **Pengfei Jiao**[4] and **Liang Yang**[5]
[1]School of Information and Intelligent Science, Donghua University, Shanghai, China
[2]Institute of Information Engineering, Chinese Academy of Sciences, Beijing, China
[3]School of Artificial Intelligence, Tianjin University, Tianjin, China
[4]School of Cyberspace, Hangzhou Dianzi University, Hangzhou, China
[5]School of Artificial Intelligence, Hebei University of Technology, Tianjin, China
gezhang@dhu.edu.cn, 2242853@mail.dhu.edu.cn, ghsun@dhu.edu.cn, xiu.fang@dhu.edu.cn, linxixun@iie.ac.cn, wangxiaobao@tju.edu.cn, pjiao@hdu.edu.cn, yangliang@vip.qq.com

1 Derivation of the PMIB Loss

Starting from Eq. (3), we first rewrite the objective as the following loss:

$$\mathcal{L}_1^{(l)} = I\Big(\mathcal{G}^{(l)};\mathcal{G}^{(l+1)} \mid \mathcal{G}_p^{(l)}\Big) - \frac{1}{\beta_1}\, I\Big(\mathcal{G}_p^{(l)};\mathcal{G}^{(l+1)}\Big). \quad \text{(S.1)}$$

Similarly, the objective in Eq. (4) can be written as:

$$\mathcal{L}_2^{(l)} = I\Big(\mathcal{G}_p^{(l)};\mathcal{G}_p^{(l+1)} \mid \mathcal{G}^{(l)}\Big) - \frac{1}{\beta_2}\, I\Big(\mathcal{G}^{(l)};\mathcal{G}_p^{(l+1)}\Big), \quad \text{(S.2)}$$

where β_1 and β_2 are trade-off parameters corresponding to the two views. We optimize the averaged joint loss:

$$\mathcal{L}_{\text{joint}} = \frac{I(\mathcal{G}^{(l)};\mathcal{G}^{(l+1)} \mid \mathcal{G}_p^{(l)}) + I(\mathcal{G}_p^{(l)};\mathcal{G}_p^{(l+1)} \mid \mathcal{G}^{(l)})}{2} - \frac{\frac{1}{\beta_1} I(\mathcal{G}_p^{(l)};\mathcal{G}^{(l+1)}) + \frac{1}{\beta_2} I(\mathcal{G}^{(l)};\mathcal{G}_p^{(l+1)})}{2}. \quad \text{(S.3)}$$

We first derive a tractable upper bound for the conditional mutual information term $I(\mathcal{G}^{(l)};\mathcal{G}^{(l+1)} \mid \mathcal{G}_p^{(l)})$. By the definition of conditional mutual information,

$$I(X;Y \mid Z) = \mathbb{E}\left[\log \frac{p(Y \mid X,Z)}{p(Y \mid Z)}\right], \quad \text{(S.4)}$$

we obtain

$$I(\mathcal{G}^{(l)};\mathcal{G}^{(l+1)} \mid \mathcal{G}_p^{(l)}) = \mathbb{E}\left[\log \frac{p(\mathcal{G}^{(l+1)} \mid \mathcal{G}^{(l)},\mathcal{G}_p^{(l)})}{p(\mathcal{G}^{(l+1)} \mid \mathcal{G}_p^{(l)})}\right]. \quad \text{(S.5)}$$

In our model, the coarsening process from $\mathcal{G}^{(l)}$ to $\mathcal{G}^{(l+1)}$ is induced by the graph structure abstraction module, yielding an implicit transition distribution $p_\theta(\mathcal{G}^{(l+1)} \mid \mathcal{G}^{(l)})$. Analogously, the graph structure abstraction module for prototype induces $q_\phi(\mathcal{G}_p^{(l+1)} \mid \mathcal{G}_p^{(l)})$. To obtain a tractable surrogate, we adopt the following approximations:

$$p(\mathcal{G}^{(l+1)} \mid \mathcal{G}^{(l)},\mathcal{G}_p^{(l)}) \approx p_\theta(\mathcal{G}^{(l+1)} \mid \mathcal{G}^{(l)}), \quad \text{(S.6)}$$

and we use the prototype transition as a reference distribution:

$$p(\mathcal{G}^{(l+1)} \mid \mathcal{G}_p^{(l)}) \approx q_\phi(\mathcal{G}_p^{(l+1)} \mid \mathcal{G}_p^{(l)}). \quad \text{(S.7)}$$

Substituting Eqs. (S.6)–(S.7) into Eq. (S.5), we obtain the following tractable upper bound:

$$I\Big(\mathcal{G}^{(l)};\mathcal{G}^{(l+1)} \,\Big|\, \mathcal{G}_p^{(l)}\Big) \lesssim D_{\text{KL}}\Big(p_\theta(\mathcal{G}^{(l+1)} \mid \mathcal{G}^{(l)}) \,\Big\|\, q_\phi(\mathcal{G}_p^{(l+1)} \mid \mathcal{G}_p^{(l)})\Big). \quad \text{(S.8)}$$

where $D_{\text{KL}}(\cdot)$ denotes the Kullback–Leibler (KL) divergence. By symmetry, we analogously have:

$$I\Big(\mathcal{G}_p^{(l)};\mathcal{G}_p^{(l+1)} \,\Big|\, \mathcal{G}^{(l)}\Big) \lesssim D_{\text{KL}}\Big(q_\phi(\mathcal{G}_p^{(l+1)} \mid \mathcal{G}_p^{(l)}) \,\Big\|\, p_\theta(\mathcal{G}^{(l+1)} \mid \mathcal{G}^{(l)})\Big). \quad \text{(S.9)}$$

Therefore, the average of the two conditional mutual information terms in Eq. (S.3) admits an upper bound by the symmetrized KL (SKL) divergence:

$$\frac{I(\mathcal{G}^{(l)};\mathcal{G}^{(l+1)} \mid \mathcal{G}_p^{(l)}) + I(\mathcal{G}_p^{(l)};\mathcal{G}_p^{(l+1)} \mid \mathcal{G}^{(l)})}{2} \lesssim D_{\text{SKL}}\Big(p_\theta(\mathcal{G}^{(l+1)} \mid \mathcal{G}^{(l)}) \,\Big\|\, q_\phi(\mathcal{G}_p^{(l+1)} \mid \mathcal{G}_p^{(l)})\Big), \quad \text{(S.10)}$$

where $D_{\text{SKL}}(p\|q) = \frac{1}{2}D_{\text{KL}}(p\|q) + \frac{1}{2}D_{\text{KL}}(q\|p)$. Next, we introduce a tractable surrogate for the preserved information part in Eq. (S.3). We aim to unify $I(\mathcal{G}_p^{(l)};\mathcal{G}^{(l+1)})$ and $I(\mathcal{G}^{(l)};\mathcal{G}_p^{(l+1)})$ into the same-granularity alignment term $I(\mathcal{G}^{(l+1)};\mathcal{G}_p^{(l+1)})$. Consider the instance-view preserved term, by the chain rule:

$$I\Big(\mathcal{G}_p^{(l)};\mathcal{G}^{(l+1)},\mathcal{G}_p^{(l+1)}\Big) = I\Big(\mathcal{G}_p^{(l)};\mathcal{G}_p^{(l+1)}\Big) + I\Big(\mathcal{G}_p^{(l)};\mathcal{G}^{(l+1)} \,\Big|\, \mathcal{G}_p^{(l+1)}\Big), \quad \text{(S.11)}$$

we assume that $\mathcal{G}_p^{(l+1)}$ is a sufficient representation for $\mathcal{G}_p^{(l)}$ with respect to aligning with $\mathcal{G}^{(l+1)}$, i.e.,

$$I(\mathcal{G}_p^{(l)};\mathcal{G}^{(l+1)} \mid \mathcal{G}_p^{(l+1)}) = 0. \quad \text{(S.12)}$$

*Corresponding author.

This assumption suggests that the information in $\mathcal{G}_p^{(l)}$ that is relevant to aligning with $\mathcal{G}^{(l+1)}$ can be represented by $\mathcal{G}_p^{(l+1)}$. Therefore, instead of directly estimating the cross-level term $I(\mathcal{G}_p^{(l)};\mathcal{G}^{(l+1)})$, we use a same-granularity surrogate between the coarsened graph and prototype:

$$I(\mathcal{G}_p^{(l)};\mathcal{G}^{(l+1)}) \;\approx\; I(\mathcal{G}_p^{(l+1)};\mathcal{G}^{(l+1)}). \tag{S.13}$$

Similarly, we obtain:

$$I(\mathcal{G}^{(l)};\mathcal{G}_p^{(l+1)}) \;\approx\; I(\mathcal{G}^{(l+1)};\mathcal{G}_p^{(l+1)}). \tag{S.14}$$

Therefore, we approximate the preserved information part in Eq. (S.3) with the following surrogate:

$$\begin{aligned}&\frac{\frac{1}{\beta_1}I(\mathcal{G}_p^{(l)};\mathcal{G}^{(l+1)}) + \frac{1}{\beta_2}I(\mathcal{G}^{(l)};\mathcal{G}_p^{(l+1)})}{2}\\ &\approx\; \frac{\frac{1}{\beta_1}+\frac{1}{\beta_2}}{2}\; I\Big(\mathcal{G}^{(l+1)};\mathcal{G}_p^{(l+1)}\Big)\,.\end{aligned} \tag{S.15}$$

Combining the SKL approximation in Eq. (S.10) with the preserved information surrogate in Eq. (S.15), we obtain:

$$\begin{aligned}\mathcal{L}_{\text{joint}}^{(l)} \;\lesssim\; & D_{\text{SKL}}\Big(p_\theta(\mathcal{G}^{(l+1)} \mid \mathcal{G}^{(l)}) \;\Big\|\; q_\phi(\mathcal{G}_p^{(l+1)} \mid \mathcal{G}_p^{(l)})\Big)\\ &- \frac{1}{2}\Big(\frac{1}{\beta_1}+\frac{1}{\beta_2}\Big) I\Big(\mathcal{G}^{(l+1)};\mathcal{G}_p^{(l+1)}\Big)\,.\end{aligned} \tag{S.16}$$

Finally, by multiplying both terms with $\beta = \frac{2\beta_1\beta_2}{\beta_1+\beta_2}$ and reparameterizing the objective, we obtain the tractable PMIB loss:

$$\begin{aligned}\mathcal{L}_{\text{PMIB}}^{(l)} = &-I\Big(\mathcal{G}^{(l+1)};\mathcal{G}_p^{(l+1)}\Big)\\ &+\;\beta\, D_{\text{SKL}}\Big(p_\theta(\mathcal{G}^{(l+1)} \mid \mathcal{G}^{(l)}) \;\Big\|\; q_\phi(\mathcal{G}_p^{(l+1)} \mid \mathcal{G}_p^{(l)})\Big)\,.\end{aligned} \tag{S.17}$$

2 Methodology Discussion

2.1 Algorithms

The overall algorithm of M-GLAD is summarized in Algorithm 1 and Algorithm 2. Algorithm 1 describes the training procedure for learning multi-granular prototypes, while Algorithm 2 presents the inference process that computes anomaly scores using the learned prototypes.

2.2 Time Complexity Analysis

We analyze the time complexity of M-GLAD in both training and inference phases. The training time complexity is $\mathcal{O}\big(L_1(|\mathcal{E}|d+|\mathcal{V}|d^2)+L_2(|\mathcal{E}|\bar{C}+|\mathcal{V}|d\bar{C})+K^2d+L_2B^2d\big)$, where L_1 is the number of GNN layers, L_2 is the number of graph structure abstraction layers, d is the feature dimension, $|\mathcal{V}|$ and $|\mathcal{E}|$ denote the total numbers of nodes and edges in a batch, $\bar{C}$ is the average number of clusters, K is the number of prototype nodes, and B is the batch size. During inference, the complexity reduces to $\mathcal{O}\big(L_1(|\mathcal{E}|d+|\mathcal{V}|d^2)+L_2(|\mathcal{E}|\bar{C}+|\mathcal{V}|d\bar{C})\big)$.

3 Detailed Experimental Setting

3.1 Datasets

We consider eight benchmark datasets in total. The statistics of the datasets are provided in Table 1. In this paper, we take "PROTEINS-F" and "IMDB-B" as the abbreviations of "PROTEINS-full" and "IMDB-BINARY", respectively.

Algorithm 1 Training of M-GLAD

Input: Training set $\mathcal{D}_{tr}$.
Parameters: Granularity L; Number of epochs E.
Output: Cached prototypes.

```
Initialize model parameters
Compute training node embeddings H and initialize graph G_p via K-means
for e = 1, 2, ..., E do
    B_1, ..., B_n ← Randomly split D_tr into batches
    for B = B_1, ..., B_n do
        {G^(l)}_{l=0}^L ← Obtain multi-granular graphs
        {z^(l)}_{l=0}^L ← Compute by graph encoder
        {G_p^(l)}_{l=0}^L ← Obtain multi-granular prototypes
        {p^(l)}_{l=0}^L ← Compute by prototype encoder
        {L_MI^(l)}_{l=0}^L ← Compute MI loss via Eq. (15)
        {L_SKL^(l)}_{l=0}^L ← Compute SKL loss via Eq. (17)
        {L^(l)}_{l=0}^L ← Compute total loss via Eq. (18)
        Update model parameters by minimizing {L^(l)}_{l=0}^L
    end for
end for
return {p^(l)}_{l=0}^L
```

Algorithm 2 Inference of M-GLAD

Input: Test set $\mathcal{D}_{te}$; Cached prototypes $\{\mathbf{p}^{(l)}\}_{l=0}^{L}$.
Output: Anomaly Score Set $\mathcal{S}$.

```
Initialize anomaly score set S ← ∅
for each graph G ∈ D_te do
    {G^(l)}_{l=0}^L ← Obtain multi-granular graphs from G
    {z^(l)}_{l=0}^L ← Compute by graph encoder
    {score^(l)(G)}_{l=0}^L ← Compute granularity-wise anomaly scores via Eq. (19)
    score(G) ← Aggregate scores via Eq. (20)
    S ← S ∪ {score(G)}
end for
return S
```

3.2 Hyper-parameters

We determine the hyper-parameters of the proposed model through empirical tuning on a held-out validation set. During tuning, we focus on a set of influential hyper-parameters and adjust them within predefined candidate ranges. Specifically, the number of training epochs is selected from $\{10, 50, 100, 200, 500\}$, and the learning rate is chosen from {1e-3,1e-4}. The depth of the graph encoder is set between 2 and 5 layers, with hidden dimensions selected from $\{32, 64, 128\}$. For the structure abstraction module, we consider 2 to 4 layers with hidden dimensions in $\{64, 128, 256\}$, and adopt commonly used GIN architecture. All hyper-parameters are fixed once selected and remain unchanged during evaluation. This tuning procedure is applied consistently across all datasets. For the baseline methods, we follow the hyper-parameter settings reported in the original implementations whenever available. When such settings are not explicitly specified, we perform light tuning on a limited

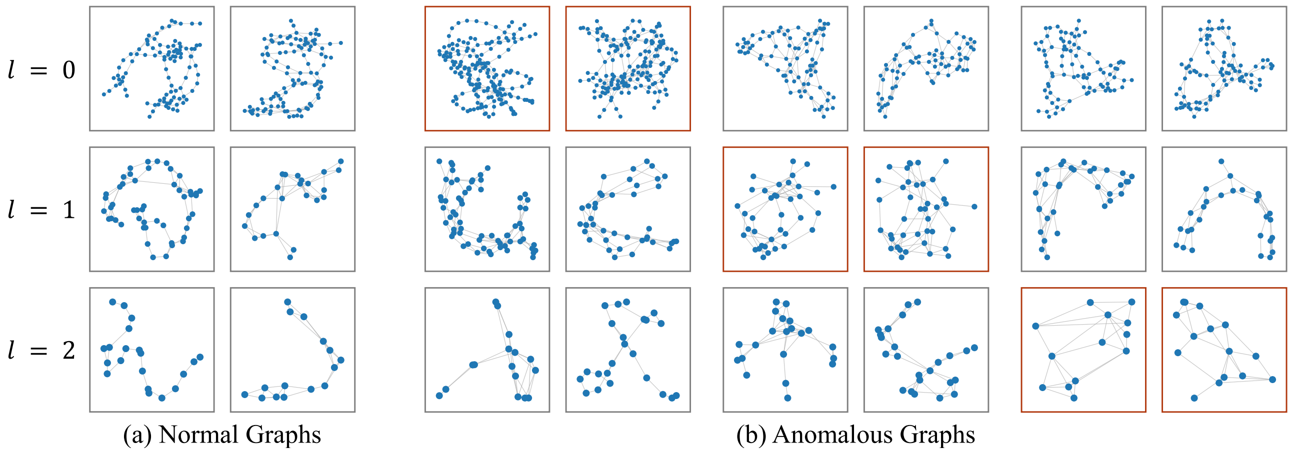

Figure 1: Multi-granular visualizations of normal and anomalous graphs on the DD dataset. Rows correspond to different granularities ($l = 0, 1, 2$). Normal graphs are shown in (a), while anomalous graphs are shown in (b). Red boxes highlight the granularity at which the anomaly signal is most salient for each instance.

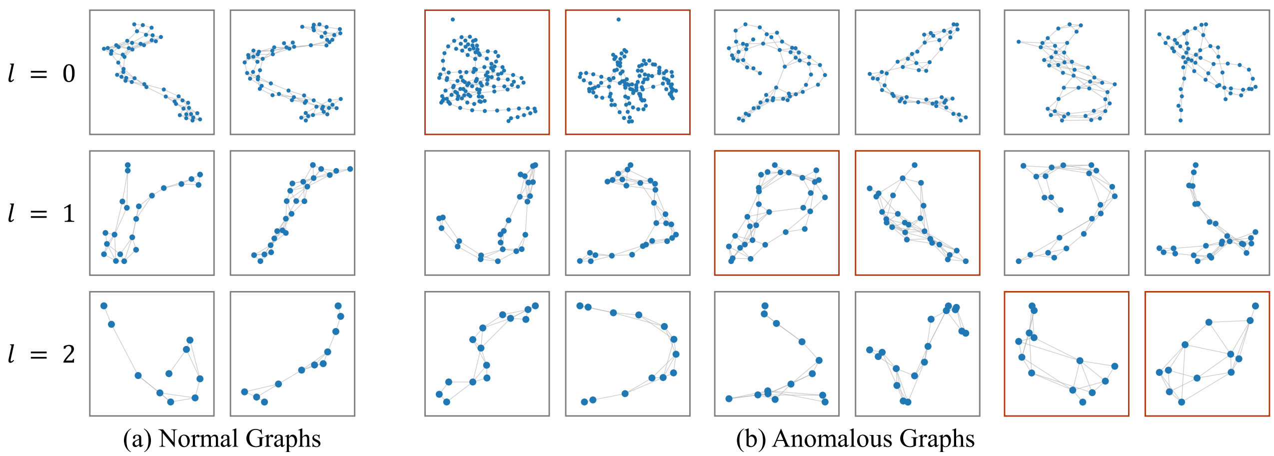

Figure 2: Multi-granular visualizations of normal and anomalous graphs on the ENZYMES dataset (same setting as Figure 1).

Dataset	#Graphs (Tr/Te)	#Nodes (avg.)	#Edges (avg.)
PROTEINS-F	360/223	39.1	72.8
ENZYMES	400/120	32.6	62.1
AIDS	1280/400	15.7	16.2
BZR	69/81	35.8	38.4
COX2	81/94	41.2	43.5
DD	390/236	284.3	715.7
MUTAG	100/38	19.5	43.6
IMDB-B	400/200	19.8	96.5

Table 1: Statistics of benchmark datasets used in our experiments.

number of sensitive parameters, such as training epochs and learning rates, to ensure stable performance without favoring any particular method.

3.3 Metrics

We evaluate model performance using AUC and AUPR.

- AUC measures the overall ranking quality of predictions by considering the trade-off between true positive and false positive rates across all thresholds.
- AUPR focuses on precision–recall behavior and provides a more informative assessment in imbalanced settings, where accurate detection of positive instances is critical and essential.

3.4 Computing Infrastructures

All experiments were conducted on a Linux server with an Intel(R) Xeon(R) Gold 5218R 20-core CPU, 64GB of RAM, and an NVIDIA RTX 4090 GPU with 24GB memory. The code is implemented using PyTorch 2.5.0, PyTorch Geometric (PyG) 2.7.0, and Python 3.10.19.

4 Descriptions of The Baseline

4.1 Global-aggregation based GLAD methods

To evaluate the effectiveness of the proposed framework for unsupervised GLAD, we compare our model with several competitive baselines, covering one-class learning, contrastive learning, knowledge distillation and subgraph modeling approaches.

- OCGIN [Zhao and Akoglu, 2020] formulates GLAD as a one-class classification problem within a GNN framework. It learns compact representations of normal graphs by enforcing them to lie within a hyperspherical region, and identifies anomalies according to their deviation from the learned normality distribution.
- OCGTL [Qiu *et al.*, 2022] extends deep one-class learning by incorporating self-supervision. Specifically, it trains multiple GNNs under complementary objectives to jointly capture global graph semantics and auxiliary structural signals, enabling more robust characterization of normal graphs.
- GLocalKD [Ma *et al.*, 2022] adopts a knowledge distillation paradigm based on random representation distillation, where a trainable GNN is optimized to predict graph- and node-level representations generated by a fixed randomly initialized network on normal graphs, and anomalies are detected via prediction discrepancies.
- GOOD-D [Liu *et al.*, 2022] leverages hierarchical contrastive learning to model normality at multiple levels. By aligning representations across node-level, graph-level, and group-level views, it detects anomalous graphs through inconsistencies between different augmented graph perspectives and their alignment with learned normal patterns.
- MUSE [Kim *et al.*, 2024] revisits graph autoencoder based model and shows that mean reconstruction error alone may be insufficient. It summarizes reconstruction errors from multiple perspectives to form graph-level features, enabling more effective anomaly detection.
- DO2HSC [Zhang *et al.*, 2024] enhances hypersphere-based anomaly detection by learning graph representations aligned with orthogonal single- or bi-hypersphere decision boundaries. It detects graph anomalies based on deviations from compact hyperspherical regions learned from normal graphs.

4.2 Subgraph-oriented GLAD methods

- SIGNET [Liu *et al.*, 2023] focuses on subgraph-level anomaly evidence by learning a cross-view mutual information objective between the original graph and its induced hypergraph representation. Anomaly scores are derived from the degree of abnormality of the extracted local substructures.
- GLADPro [Yang *et al.*, 2025] introduces a prototype-based framework to improve the interpretability of GLAD. It represents normal graph distributions using a set of learned prototypes. Two variants are considered: one produces instance-specific explanations, and the other learns global prototypes and heuristically associates them with representative subgraphs.

5 Further Supplementary of Visualizations

This section presents additional qualitative visualizations on the DD and ENZYMES datasets. Figure 1 and Figure 2 summarize the observations for DD and ENZYMES, respectively. Each figure is arranged by graph granularity, with rows corresponding to $l = 0, 1, 2$. Within each row, (a) shows normal graphs and (b) shows anomalous graphs. Red boxes mark the granularity where the anomaly is most visually evident for that instance. Across both datasets, the examples suggest that anomalies can be clearly manifested at certain granularities, while remaining weak or visually inconspicuous at others. At granularity $l = 0$, anomalous graphs often differ subtly from normal ones, typically through slightly denser local connections that may be difficult to distinguish by visual inspection. At granularity $l = 1$, differences begin to exhibit more organized structural patterns: some anomalous graphs form noticeably compact regions or show uneven connectivity, although the overall structure may still look broadly similar. At granularity $l = 2$, anomalous graphs may exhibit distinctive structural configurations, such as ring-like or strongly cyclic patterns, which are rarely observed in normal graphs. Overall, these examples indicate that an anomaly is not always tied to one specific local irregularity. Instead, abnormal patterns may remain weak or ambiguous at certain granularities and become substantially more distinguishable at others. This observation supports modeling graphs across multiple granularities for unsupervised GLAD.

References

[Kim *et al.*, 2024] Sunwoo Kim, Soo Yong Lee, Fanchen Bu, Shinhwan Kang, Kyungho Kim, Jaemin Yoo, and Kijung Shin. Rethinking reconstruction-based graph-level anomaly detection: Limitations and a simple remedy. In *NeurIPS*, 2024.

[Liu *et al.*, 2022] Yixin Liu, Kaize Ding, Huan Liu, and Shirui Pan. Good-d: On unsupervised graph out-of-distribution detection. In *WSDM*, 2022.

[Liu *et al.*, 2023] Yixin Liu, Kaize Ding, Qinghua Lu, Fuyi Li, Leo Yu Zhang, and Shirui Pan. Towards self-interpretable graph-level anomaly detection. *NeurIPS*, 36:8975–8987, 2023.

[Ma *et al.*, 2022] Rongrong Ma, Guansong Pang, Ling Chen, and Anton Van Den Hengel. Deep graph-level anomaly detection by glocal knowledge distillation. In *WSDM*, 2022.

[Qiu *et al.*, 2022] Chen Qiu, M. Kloft, Stephan Mandt, and Maja R. Rudolph. Raising the bar in graph-level anomaly detection. In *IJCAI*, 2022.

[Yang *et al.*, 2025] Zhenyu Yang, Ge Zhang, Jia Wu, Jian Yang, Shan Xue, Amin Beheshti, Hao Peng, and Quan Z Sheng. Global interpretable graph-level anomaly detection via prototype. In *KDD*, 2025.

[Zhang *et al.*, 2024] Yunhe Zhang, Yan Sun, Jinyu Cai, and Jicong Fan. Deep orthogonal hypersphere compression for anomaly detection. In *ICLR*, 2024.

[Zhao and Akoglu, 2020] Lingxiao Zhao and Leman Akoglu. On using classification datasets to evaluate graph outlier detection: Peculiar observations and new insights. *Big Data*, 11:151 – 180, 2020.